

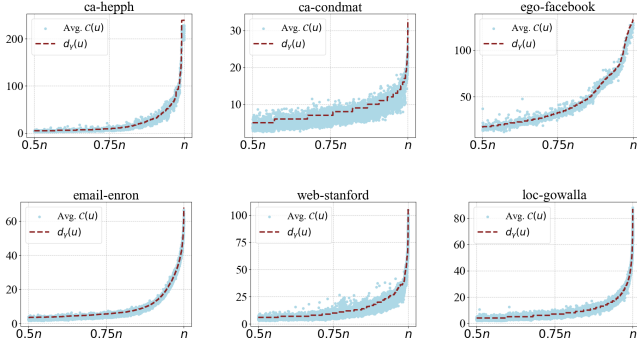


Figure 9: Average number of credits over 10 runs. The x -axis reports vertices sorted by γ -degree, while the y -axis shows the average number of credits. Results are shown for static graphs. This representation highlights that the estimator preserves the relative ordering of vertices.

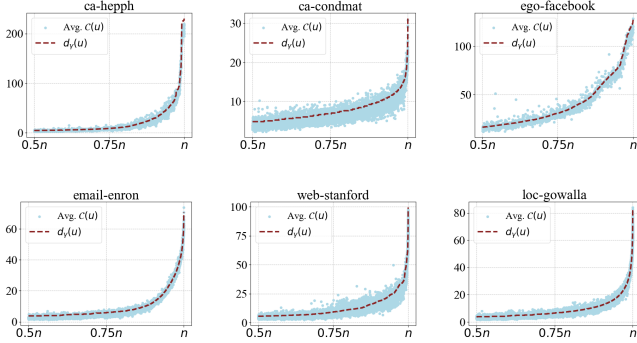


Figure 10: Average number of credits over 10 runs. The x -axis reports vertices sorted by γ -degree, while the y -axis shows the average number of credits. Results are shown for dynamic synthetic sequences with $p = 0.1$. This representation highlights that the estimator preserves the relative ordering of vertices.

A DEFERRED PROOFS

A.1 Proof of Lemma 3.1

PROOF. Let $N'(x)$ be neighborhood of x after the deletion. Since only the edge (x, y) is removed, we have $N'(x) = N(x) \setminus \{y\}$, so $|N'(x)| = |N(x)| - 1$. For any $w \notin \{x, y\}$, the neighborhood

$N'(w) = N(w)$ remains unchanged, hence $|N'(w)| = |N(w)|$. We distinguish two cases based on whether $w \in N(y)$.

Case 1: $w \in N(x) \cap N(y)$. Since $w \in N(y)$, we have $y \in N(w) \cap N(x)$. After removing (x, y) , $y \notin N'(x)$, so $|N'(w) \cap N'(x)| = |N(w) \cap N(x)| - 1$. Therefore:

$$\tau'(x, w) = \frac{|N'(w) \cap N'(x)|}{|N'(x)|} = \frac{|N(w) \cap N(x)| - 1}{|N(x)| - 1} = \frac{\tau(x, w) |N(x)| - 1}{|N(x)| - 1},$$

$$\tau'(w, x) = \frac{|N'(w) \cap N'(x)|}{|N'(w)|} = \frac{|N(w) \cap N(x)| - 1}{|N(w)|} = \tau(w, x) - \frac{1}{|N(w)|}.$$

Case 2: $w \in N(x) \setminus N(y)$. Since $w \notin N(y)$, y does not contribute to $N(w) \cap N(x)$, so $|N'(w) \cap N'(x)| = |N(w) \cap N(x)|$. Therefore:

$$\tau'(x, w) = \frac{|N'(w) \cap N'(x)|}{|N'(x)|} = \frac{|N(w) \cap N(x)|}{|N(x)| - 1} = \frac{\tau(x, w) |N(x)|}{|N(x)| - 1},$$

$$\tau'(w, x) = \frac{|N'(w) \cap N'(x)|}{|N'(w)|} = \frac{|N(w) \cap N(x)|}{|N(w)|} = \tau(w, x).$$

□

A.2 Proof of Lemma 3.2

PROOF. Let $N'(x)$ be neighborhood of x after the insertion. Since only the edge (x, y) is added, we have $N'(x) = N(x) \cup \{y\}$, so $|N'(x)| = |N(x)| + 1$. For any $w \notin \{x, y\}$, the neighborhood $N'(w) = N(w)$ remains unchanged, hence $|N'(w)| = |N(w)|$. We distinguish two cases based on whether $w \in N(y)$.

Case 1: $w \in N(x) \cap N(y)$. Since $w \in N(y)$, we have $y \in N(w) \cap N'(x)$. After adding (x, y) , $y \in N'(x)$, so $|N'(w) \cap N'(x)| = |N(w) \cap N(x)| + 1$. Therefore:

$$\tau'(x, w) = \frac{|N'(w) \cap N'(x)|}{|N'(x)|} = \frac{|N(w) \cap N(x)| + 1}{|N(x)| + 1} = \frac{\tau(x, w) |N(x)| + 1}{|N(x)| + 1},$$

$$\tau'(w, x) = \frac{|N'(w) \cap N'(x)|}{|N'(w)|} = \frac{|N(w) \cap N(x)| + 1}{|N(w)|} = \tau(w, x) + \frac{1}{|N(w)|}.$$

Case 2: $w \in N(x) \setminus N(y)$. Since $w \notin N(y)$, y does not contribute to $N(w) \cap N(x)$, so $|N'(w) \cap N'(x)| = |N(w) \cap N(x)|$. Therefore:

$$\tau'(x, w) = \frac{|N'(w) \cap N'(x)|}{|N'(x)|} = \frac{|N(w) \cap N(x)|}{|N(x)| + 1} = \frac{\tau(x, w) |N(x)|}{|N(x)| + 1},$$

$$\tau'(w, x) = \frac{|N'(w) \cap N'(x)|}{|N'(w)|} = \frac{|N(w) \cap N(x)|}{|N(w)|} = \tau(w, x).$$

□

B FURTHER EXPERIMENTS

B.1 γ -degree estimation

Figures 9 and 10 present the experimental results for the γ -degree estimation using the credit-based estimator. Specifically, Figure 9 illustrates the performance on insertion-only sequences, while Figure 10 focuses on fully dynamic sequences. In both figures, the sequences are generated according to the procedure explained in Section 6.

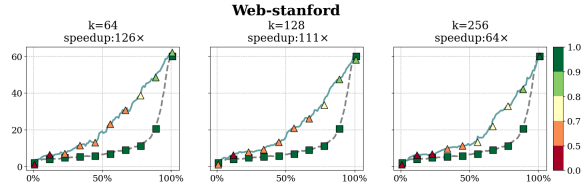


Figure 11: Quality of the solutions obtained using the random permutation sequence with $p = 0.1$ using various value of k

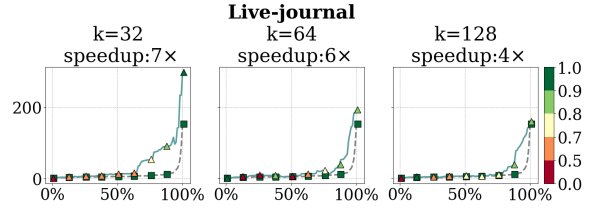


Figure 12: Quality of the solutions obtained using the incremental random permutation sequence using various value of k

B.2 Comparison with baseline

In this section, we extend the experimental evaluation on synthetic datasets from Section 6.4, demonstrating the performance of our solution as the min-hash signature size k varies. Experimental results for web-stanford and live-journal are shown in Figure 11 and Figure 12, respectively.